University of Central Punjab -FOIT
COAL Fall 2025
Quiz 2

1. The 8-bit number CL=0b10110111 has its lower nibble (4 bits) representing sensor data. Extract the lower nibble and convert it to decimal.

Solution

Given: $AL = 0b10110111$

1. **Extract the Lower Nibble**:
   - The lower nibble of an 8-bit number consists of the least significant 4 bits (rightmost 4 bits).
   - To extract these bits, use a bitwise AND operation with the mask $0b00001111$ (decimal 15):

   $$AL \wedge 0b00001111 = 0b10110111 \wedge 0b00001111 = 0b00000111$$

   - The lower nibble is $0b00000111$.
2. **Convert to Decimal**:
   - $0b00000111 = 7 \text{ (in decimal)}$.

Final Answer:

The lower nibble extracted is **7** (in decimal).

2. Perform a left shift by 2 positions on DL=0b00111010. What is the new binary value, and what happens if there is overflow?

1. **Perform the Left Shift**:
   In a left shift operation, the bits in the number are shifted to the left, and zeros are filled in from the right.
   - Original value: $AL = 0b00111010$.
   - Shift left by 2 positions:

   $$0b00111010 \rightarrow 0b11101000$$

   New binary value: $0b11101000$.
2. **Overflow Check**:
   - An 8-bit register can hold values from 0 to 255 ($0b00000000$ to $0b11111111$).
   - If bits shift beyond the 8th position (MSB), they are **lost**, causing overflow.
   - In this case, no bits are shifted out beyond the MSB, so **no overflow occurs**.

3. Apply a series of operations on AL=0b01100111:
a) Perform a rotate left by 2 positions.
b) Perform a right arithmetic shift by 3 positions.
c) Rotate right by 1 position.
Provide the final result in both binary and decimal. Perform binary to decimal conversion at each step.

a) Rotate left by 2 positions:

Binary: 10011101, Decimal: 157

b) Right arithmetic shift by 3 positions:

Binary: 11110011, Decimal: -13

c) Rotate right by 1 position:

Binary: 11111001, Decimal: -7

157
243
249

4. Consider the following lines of code and give the value stored in CX after each instruction.
Note: CF = 1; CX = 10101100

SHL CX
XOR CX, 11110000
RCR CX

Initial	10101100	1
SHL CX	01011000	1
XOR CX, 11110000	10101000	1
RCR CX	11010100	0